Assignment 1

Analyze Canadian health & income data.
Dataset: `Canadian_Health_Survey_Sample.csv`

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

Part A – Understanding the Data

A1) Load CSV

```
df = pd.read_csv("../assets/Canadian_Health_Survey_Sample.csv")
```

A2) Information

```
df.head()
```

	Province	Gender	Age	Income	BMI	PhysicalActivity	Smoking	SelfRatedHealth	StressLevel
0	Ontario	female	59	80572.0	25.7	3.8	yes	Good	6.0
1	New Brunswick	Male	38	125739.0	24.2	3.4	No	Good	9.0
2	Nova Scotia	M	31	55997.0	35.7	NaN	No	Excellent	NaN
3	Saskatchewan	Male	30	75947.0	28.3	3.9	No	Excellent	10.0
4	Saskatchewan	female	79	113966.0	31.2	1.6	No	Fair	6.0

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 850 entries, 0 to 849
Data columns (total 9 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   Province          850 non-null    object
 1   Gender            850 non-null    object
 2   Age               850 non-null    int64
 3   Income            816 non-null    float64
```

```
 4   BMI               808 non-null    float64
 5   PhysicalActivity  825 non-null    float64
 6   Smoking           850 non-null    object
 7   SelfRatedHealth   850 non-null    object
 8   StressLevel       825 non-null    float64
dtypes: float64(4), int64(1), object(4)
memory usage: 59.9+ KB
```

```
df.describe(include='all')
```

	Province	Gender	Age	Income	BMI	PhysicalActivity	Smoking	SelfRatedHealth	StressLevel
count	850	850	850.000000	816.000000	808.000000	825.000000	850	850	825.000000
unique	10	6	NaN	NaN	NaN	NaN	6	5	NaN
top	Quebec	Male	NaN	NaN	NaN	NaN	No	Good	NaN
freq	100	230	NaN	NaN	NaN	NaN	552	258	NaN
mean	NaN	NaN	49.517647	77461.917892	27.488861	3.511394	NaN	NaN	5.294545
std	NaN	NaN	18.192604	22497.833440	5.064344	1.925825	NaN	NaN	2.121156
min	NaN	NaN	18.000000	5000.000000	13.500000	0.000000	NaN	NaN	1.000000
25%	NaN	NaN	33.000000	63438.000000	24.000000	2.100000	NaN	NaN	4.000000
50%	NaN	NaN	49.000000	78130.000000	27.300000	3.500000	NaN	NaN	5.000000
75%	NaN	NaN	66.000000	90529.500000	30.900000	4.700000	NaN	NaN	7.000000
max	NaN	NaN	80.000000	250000.000000	60.000000	10.400000	NaN	NaN	10.000000

A3) Attributes

Quantitative Attribute: Age - Numerical and ordered.
Age is just a number. Age is ordered as it has a size to it, 15 years old is younger than 20 years old.
Qualitative Attribute: Province - Categorical and unordered.
Province is a string category. Province does not have a size, it does not make sense to say one pronvicne is greater than another.

Part B – Cleaning & Structuring

B1) Normalization

Standardize naming for values.

```
# Check all existing values for genders
df["Gender"].unique()
```

```
array(['female', 'Male', 'M', 'F', 'Female', 'male'], dtype=object)
```

```
# Normalize
gender_map = {
    "male": "Male",
    "m": "Male",
    "female": "Female",
    "f": "Female"
}

df["Gender"] = df["Gender"].str.lower().map(gender_map)

df["Gender"].unique()
```

```
array(['Female', 'Male'], dtype=object)
```

```
# Check all existing values for smoking
df["Smoking"].unique()
```

```
array(['yes', 'No', 'Y', 'Yes', 'N', 'no'], dtype=object)
```

```
# Normalize
smoking_map = {
    "yes": "Yes",
    "y": "Yes",
    "no": "No",
    "n": "No"
}

df["Smoking"] = df["Smoking"].str.lower().map(smoking_map)

df["Smoking"].unique()
```

```
array(['Yes', 'No'], dtype=object)
```

B2) Dropna / Imputation

Remove or replace 'nan' values.

Variable	Approach	Reason
BMI	Median imputation	BMI is continuous and can have outliers → median is robust.
Income	Median imputation	Income is usually skewed → median better than mean.
StressLevel	Median imputation	Ordinal numeric (1–10), no huge skew, median is safe.
PhysicalActivity	Median imputation	Depends on % missing, if few rows missing, median is fine.

```
# Check nan counts per column
df[["BMI", "Income", "StressLevel", "PhysicalActivity"]].isna().sum()
```

```
BMI                 42
Income              34
StressLevel         25
PhysicalActivity    25
dtype: int64
```

```
# Handle nan values
df["BMI"] = df["BMI"].fillna(df["BMI"].median())
df["Income"] = df["Income"].fillna(df["Income"].median())
df["StressLevel"] = df["StressLevel"].fillna(df["StressLevel"].median())
df["PhysicalActivity"] = df["PhysicalActivity"].fillna(df["PhysicalActivity"].median())

df[["BMI", "Income", "StressLevel", "PhysicalActivity"]].isna().sum()
```

```
BMI                 0
Income              0
StressLevel         0
PhysicalActivity    0
dtype: int64
```

B3) Data Typing

Cast values to preferred data type.

```
# Check exiting data types
df.dtypes
```

```
Province             object
Gender               object
```

```
Age                   int64
Income              float64
BMI                 float64
PhysicalActivity    float64
Smoking              object
SelfRatedHealth      object
StressLevel         float64
dtype: object
```

```
# Enforce data types
df["Age"] = df["Age"].astype(int)
df["Income"] = df["Income"].astype(float)
df["BMI"] = df["BMI"].astype(float)
df["StressLevel"] = df["StressLevel"].astype(int)

df.dtypes
```

```
Province             object
Gender               object
Age                   int64
Income              float64
BMI                 float64
PhysicalActivity    float64
Smoking              object
SelfRatedHealth      object
StressLevel           int64
dtype: object
```

B4) Derivation

Get derived values from exisitng data.

BMI Category	BMI Range
Underweight	< 18.5
Normal	18.5–24.9
Overweight	25–29.9
Obese	≥ 30

```
# Create derived columns
df["BMI_Category"] = pd.cut(
```

```
    df["BMI"],
    bins=[0, 18.5, 25, 30, float('inf')],
    labels=["Underweight", "Normal", "Overweight", "Obese"],
    right=False
)

df["IncomeBracket"] = pd.cut(
    df["Income"],
    bins=[0, 50000, 80000, 110000, float('inf')],
    labels=["<50k", "50-80k", "80-110k", ">110k"],
    right=False
)

df_cleaned = df.copy(deep=True)

df[["BMI", "BMI_Category", "Income", "IncomeBracket"]].head()
```

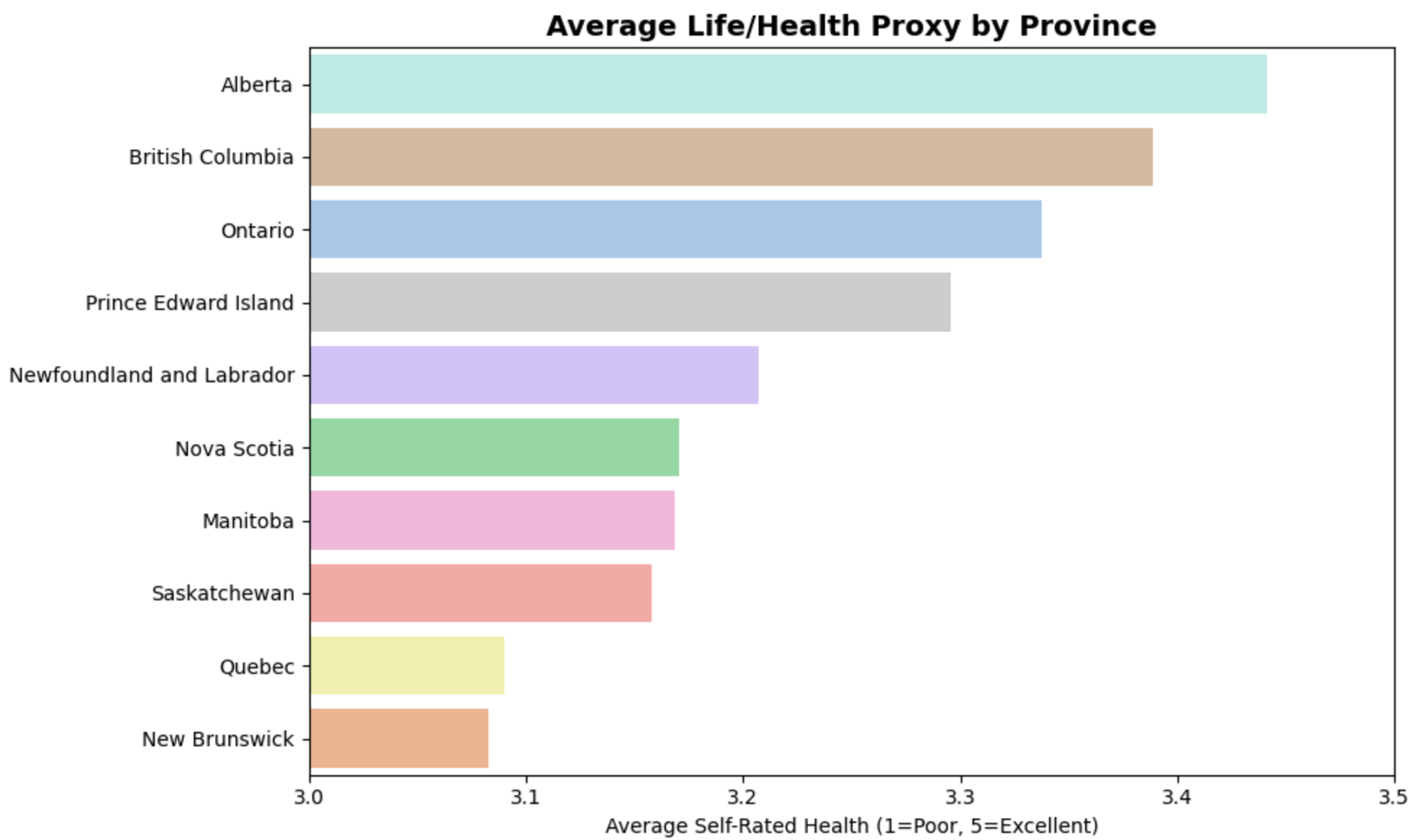
	BMI	BMI_Category	Income	IncomeBracket
0	25.7	Overweight	80572.0	80-110k
1	24.2	Normal	125739.0	>110k
2	35.7	Obese	55997.0	50-80k
3	28.3	Overweight	75947.0	50-80k
4	31.2	Obese	113966.0	>110k

Part C – Visualization

C1) Bar Chart

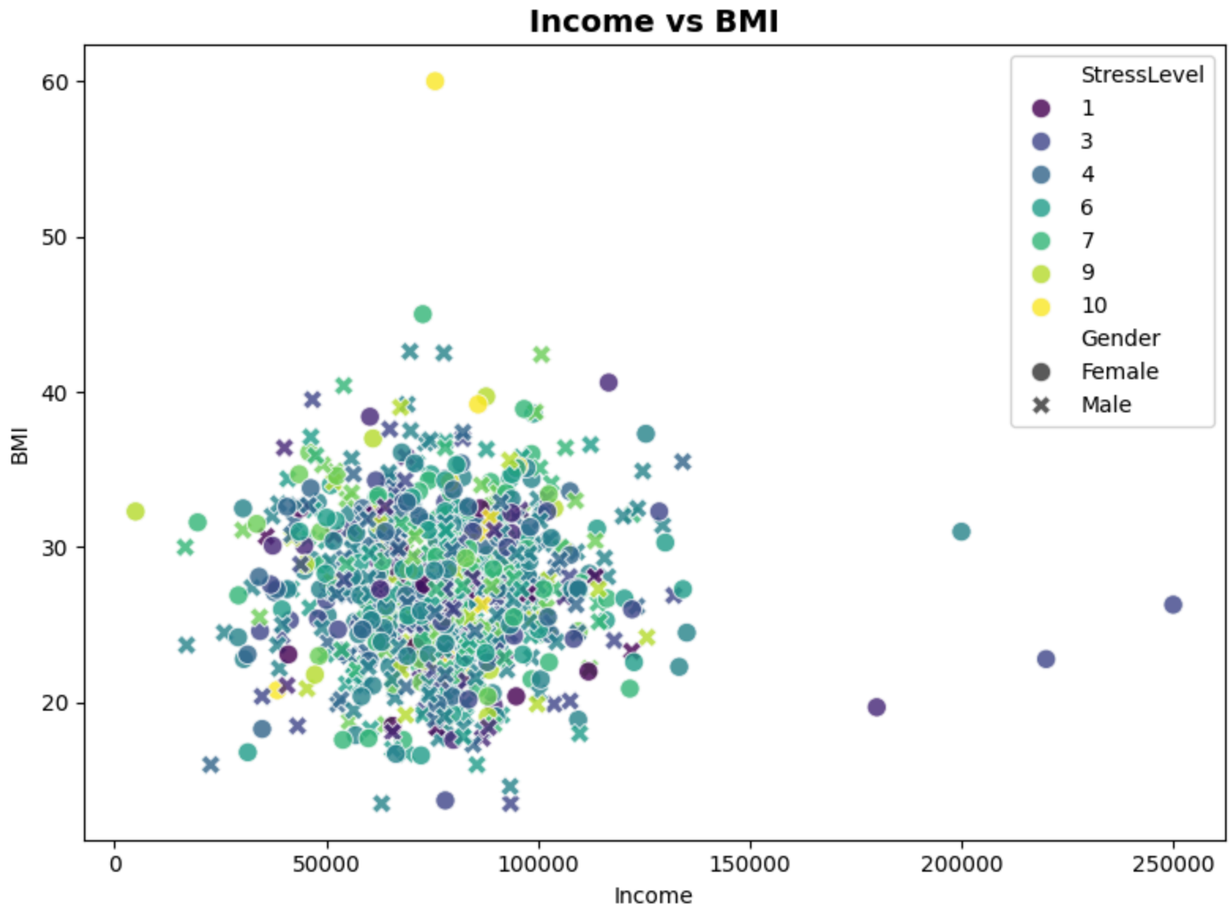
```
# Bar chart
health_map = {
    "Poor": 1,
    "Fair": 2,
    "Good": 3,
    "Very Good": 4,
    "Excellent": 5
}

df["SelfRatedHealthNumber"] = df["SelfRatedHealth"].map(health_map)
order = df.groupby("Province")["SelfRatedHealthNumber"].mean().sort_values(ascending=False).index

plt.figure(figsize=(10,6))
sns.barplot(
    data=df,
```

```
    x="SelfRatedHealthNumber",
    y="Province",
    order=order,
    orient="h",
    errorbar=None,
    hue="Province",
    palette="pastel"
)

plt.title("Average Life/Health Proxy by Province", fontsize=14, fontweight='bold')
plt.xlabel("Average Self-Rated Health (1=Poor, 5=Excellent)")
plt.ylabel("")

plt.xlim(3, 3.5)            # zoom y-axis to focus between 3.0 to 3.5 where all values lie
plt.tight_layout()
plt.show()
```



C2) Scatterplot

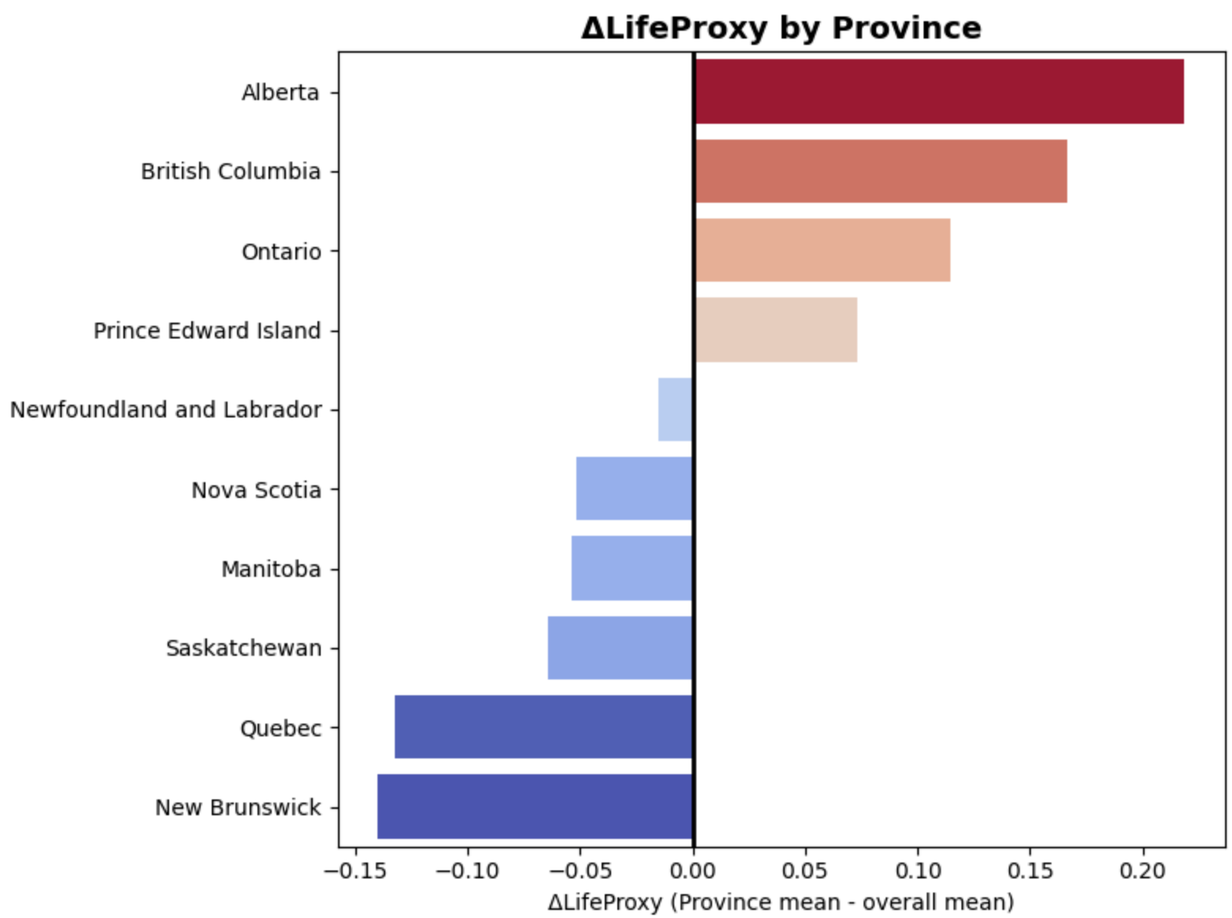
```
plt.figure(figsize=(8,6))
sns.scatterplot(
    data=df,
    x="Income",
    y="BMI",
    style="Gender",
    hue="StressLevel",
    palette="viridis",
    s=80,
    alpha=0.8
)

plt.title("Income vs BMI", fontsize=14, fontweight='bold')

plt.tight_layout()
plt.show()
```



C3) Diverging Visualization

```
overall_mean = df["SelfRatedHealthNumber"].mean()
delta_life = df.groupby("Province")["SelfRatedHealthNumber"].mean() - overall_mean
# df["ΔLifeProxy"] = df.groupby("Province")["SelfRatedHealthNumber"].transform("mean") - overall_mean       # not useful, but interesting

plt.figure(figsize=(8,6))
sns.barplot(
    x=delta_life.values,
    y=delta_life.index,
    order=delta_life.sort_values(ascending=False).index,
    orient="h",
    hue=delta_life.values,
    palette="coolwarm",
    legend=False
)
```

```
plt.title("ΔLifeProxy by Province", fontsize=14, fontweight='bold')
plt.xlabel("ΔLifeProxy (Province mean − overall mean)")
plt.ylabel("")

plt.axvline(0, color="black", linewidth=2)
plt.tight_layout()
plt.show()
```



C4) Design Choices

1. **Types of data**

- Quantitative/numerical continuous: C1 average self-rated health, C2 BMI, income
- Quantitative/numerical discrete: C2 stress level, C3 delta life proxy (by province)
- Qualitative/categorical unordered: C1 provinces, C2 genders, C3 provinces

1. **Color Scale**

- Colors - different color to represent categorical data. E.g. the colors in the bar chart in C1 are used to represent different provinces.
- Hues - a saturation gradient to represent numerical data. E.g. the blue to red hue in C3 is used to represent the negative to positive delta life proxy.

2. **Shapes**

- different shapes to represent categorical data E.g. different shapes are used along side color in C2 to represent male and female.

3. **Cognitive Load Theory (CLT)**

- Redundancy - no labels or other data are repeated in the charts.
- Coherence - no unnecessary information are presented in the charts.
- Signaling - the title of the charts are bold cased.

4. **Others**

- Descending Orders - uses descending order of values for unordered indexes.
- Horizontal Bars - prevent overlapping index values.
- Remove Obvious Labels - e.g. Province

Part D – Interpretation

D1) Discusssion

C1): The chart provide statistical data for how citizen of each Canadian province rate their own health. It does not suggest any explanation as to why the differences exist. Using this chart, we can infer that Western provinces (e.g., Alberta, British Columbia, Ontario) report higher average health, while Atlantic provinces and Quebec tend to report lower averages.
C2): This chart displays no strong linear relationship between income and BMI. It does show a central clustering around income of ~$40,000–$100,000 and BMI of ~20–35. The chart provides no significant trend for stress leve and gender differences.
C3): The chart compares values in C1) with the overall mean, showing Alberta, British Columbia, and Ontario have the largest positive deviations above the national average. Additonally, this chart shows French speaking regions have much lower average health proxy than the rest.

D2) Limitations

C1&C3): The chart does not present any correlation due to provinces being an unordered data. By presenting additional data via color coding (like C2) or other means, it can provide an explanation to why differences between provinces exist.
C2): The dataset is skewed toward average income and BMI values, while higher and lower income and BMI groups are underrepresented. As a result, this dataset reflects the distribution of the general population more so than the correlation between the variables. A more balanced sample would improve the analysis. In addition, the color and shape coding failed to present data effectively. A better approach is to seperate those categories into its own graphs for a much more effective representation of the data.

Submission

Export

```
# Export cleaned CSV
df_cleaned.to_csv("../outputs/cleaned.csv", index=False)
df_cleaned
```

	Province	Gender	Age	Income	BMI	PhysicalActivity	Smoking	SelfRatedHealth	StressLevel	BMI_Category	IncomeBracket
0	Ontario	Female	59	80572.0	25.7	3.8	Yes	Good	6	Overweight	80-110k
1	New Brunswick	Male	38	125739.0	24.2	3.4	No	Good	9	Normal	>110k
2	Nova Scotia	Male	31	55997.0	35.7	3.5	No	Excellent	5	Obese	50-80k
3	Saskatchewan	Male	30	75947.0	28.3	3.9	No	Excellent	10	Overweight	50-80k
4	Saskatchewan	Female	79	113966.0	31.2	1.6	No	Fair	6	Obese	>110k
...	...	...	...	...	...	...	...	...	...	...	...
845	Ontario	Female	68	102355.0	25.5	6.3	No	Fair	4	Overweight	80-110k
846	Quebec	Female	46	78130.0	13.7	3.0	Yes	Poor	3	Underweight	50-80k
847	Ontario	Female	78	108434.0	24.1	3.5	Yes	Good	3	Normal	80-110k
848	Newfoundland and Labrador	Male	18	85176.0	24.6	5.3	No	Very Good	4	Normal	80-110k
849	Nova Scotia	Male	63	96105.0	27.3	5.7	No	Excellent	6	Overweight	80-110k

850 rows × 11 columns